



Piri Reis University
Faculty of Economics and Administrative Sciences
Management Information Systems Department

GRADUATION THESIS - YBS421

**Real - Time Commercial Vessels Detection With Using
Yolov8**

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Istanbul (June), 2026

DECLARATION

I hereby declare that this study, titled *"Ship Detection With Using Yolov8"*, is my own original work and has not been submitted previously, in whole or in part, for the award of any degree or diploma at any other institution.

All sources and materials used in the preparation of this thesis have been properly acknowledged and cited.

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ABSTRACT

Real Time Commercial Vessels Detection With Using Yolov8

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Bachelor's Thesis in Management Information Systems

Project Supervisor: Dr. Volkan USLAN

June, 2026

In this thesis, we include the Management Information Systems field and the maritime activities of our school in our project that is a ship identification project was carried out by completing artificial intelligence model training with the Yolov8 algorithm. The main aim of the project is to integrate artificial intelligence systems into the maritime sector and with that help future studies. Due to that, our project's datasets were gathered from the Roboflow platform, and our model was trained (70%), validated (20%), and tested (10%) by taking the dataset using API Key in the Google Colab environment.

The outputs are confusion matrix, normalized confusion matrix, f1 curve, PR curve, R curve, and visual graphs that we analyzed. With that we evaluated the performance of our model using these methods. The result we achieved in this project was a success rate above 80% of the mAP50(B) score. We tested this in real-time with the model under different conditions, including images and videos that not present in the dataset, and the results showed that our model successfully identified and classified ship types.

In conclusion, This thesis is a study in the fields of Management Information Systems and Maritime, research on real-time ship identification using YOVO8 technology, focused on findings we find and expected to help future technological investments or operations. This thesis was prepared at the undergraduate study and satisfy the requirements and ethical standards expected from a graduation project.

Keywords: Roboflow,API,Google Colab, Yolov8, data analysis, results, literature review,AI,Metrics,mAP,Ship Detection,Computer Vision System,Classification

ÖZET

Yolov8 İle Gerçek Zamanlı Ticari Gemileri Tanımlama

Deniz İlkiz
Yönetim Bilişim Sistemleri Bitirme Projesi

Proje Danışmanı: Dr. Volkan USLAN

Haziran, 2026

Bu bitirme projesinde, Yönetim Bilişim Sistemleri alanı ve okulumuzun sahip olduğu denizcilik faaliyetlerini göz önüne alarak Yolov8 algoritması ile yapay zekâ modeli eğitimini tamamlayarak bir gemi tanımlama projesi gerçekleştirilmiştir. Projenin temel amacı, denizcilik sektörüne yapay zekâ sistemini entegre edip gelecek çalışmalara öncü olmak. Bu doğrultu da projemizin veri setlerini Roboflow platformundan sağlayıp API ile Google Colab ortamında veri setini çağırıp modelimiz eğitilmiş (%70), doğrulanmış (%20) ve test (%10) edilmiştir.

Çıktılar; confusion matrix, confusion matrix normalized, f1 curve, PR curve, R curve ve sonuç olarak görsel grafikler ile sağlanmış ve yorumlama yapılmıştır. Modelimizin performansını bu sayede değerlendirdik. Sonuç, mAP50(B) skorunun %80 üzeri bir başarı elde etti. Bu durumu gerçek zamanlı olarak modeli farklı koşullar altında veri setinde olmayan görsel ve videolar ile test ettik ve elde ettiğimiz sonuç ise modelimiz gemi türlerini başarılı bir şekilde tanımladı ve sınıflandırdı.

Bu proje, Yönetim Bilişim Sistemleri ve Denizcilik alanına elde edilen bulgular ile gelecekte yapılması beklenen teknolojik yatırım veya operasyonlara gerçek zamanlı yolov8 kullanılarak gemi tanımlamaya yönelik bir çalışmadır. Çalışma, lisans düzeyinde hazırlanmıştır ve bitirme projesinden beklenen şartları ve etikliği karşılamıştır.

Anahtar Kelimeler: Roboflow, API, Google Colab, Yolov8, Veri Analizi, Yapay Zeka Model Eğitimi, Gemi Tanımlama, Etiketlendirme, Bilgisayar Görü Sistemi, Grafik Analizi, Sınıflandırma

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ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Networks
YOLO	You Only Look Once
GPU	Graphics Processing Unit
GPU T4	Graphics Processing Unit Tesla 4
mAP	Mean Average Precision
FPA	False Positive Analysis
FPS	Frames Per Second
Iot	Internet Of Things

1. INTRODUCTION

My project focus on ship detection in maritime sector. This sector globalization process and steady increase in international trade volume made maritime transport and port logistics most critical supply chain. Furthermore 80% of world trade is carried by ships which increasing cargo process capacity and operations[1].

So, traditional port management, ship detection approaches are insufficient in time management and cost optimization cause container traffic happens. In this topic real time ship detection can aim technology port ecosystem integration with using Iot[2]. Our project aim is to focus using computer vision and yolov8 model object detection system for detect and classification systems in maritime security or port management etc. Because traditional ship detection system based on manual observation techniques. As a result new development must be happen[3].

As a solution smart port concept come to surface this term refers automated port system create less human work and resources functionality. Due to ship trade speed and management in port areas are so important so that is why AI enhances solution can solve this problem [4].

Search are show us that smart port increase efficiency operations. The technology used for this Iot, AI, Big Data, Sensors. Due to high cost and implementation price create a challenge for optimize maritime industry[5].

In my project, we focused to solve this problem so we come to conclusion to use Roboflow and made ship detection with yolov8. Additionally, this system create less cost in port because cameras is our eyes to detect ships. Our dataset 700 labeled ship images done by using Roboflow and used for train a model in Google Colab environment with YOLOv8 architecture 50 epochs for detect and classification vessels. For control performance objective metrics utilized. Such as, FPS for testing real time video records. After that mAP, recall, precision standard object detection metrics. With this project we want to help fill the gap in the literature.

Thesis structure focus 5 main part these are introduction, literature review, methods and materials, results and discussion, conclusion.

2. LITERATURE REVIEW

In my graduation project Object Detection Technologies used this is important for computer vision. This is image processing techniques rely on features, edge detection, shape, color analysis. However, before deep learning development happen this methods failed to achieve good results because of different environment condition or backgrounds. When CNNs come object detection performance improved and give better accuracy.

YOLO is most popular object detection method and made by Joseph Redmon in 2015. Furthermore, yolo method divide image data into a grid system and each cell focused on object within complete the process single pass. This achieve high FPS value. YOLO architecture has 12 model these are called, YOLO, YOLOv2, YOLOv3, YOLOv4, Ultralytics YOLOv5, YOLOv6, YOLOv7, Ultralytics, YOLOv8, YOLOv9 YOLOv10, Ultralytics Yolo11, Ultralytics YOLO26.

Ultralytics YOLOv8 used in my ship detection project this model released in 2023. YOLOv8 feature focused on cutting-edge performance in terms of accuracy and speed. For scalability offers 5 different parameter size Nano, small, medium, large and extra large to use in project adapted to devices or cloud servers. YOLOv8 is different because abandoned the anchor based approach that used versions and uses an anchor free detection approach. Anchor free system has a improved bounding box accuracy and focus objects center. So, this model offer high learning capacity.

Literature review show us that mostly ship detection projects use data augmentation due to less dataset findings and model's background noise that is why data augemantation techniques such as, blur image, 9 rotate, flip, brightness, noise. This help model from overfitting and get better performance metrics score.

All before that system come ship detection were done by using background subtractions, edge detection and hand made feature HOG, SIFT images from fixed cameras. These systems is classical machine learning algorithms. In summary, inside the all of these methods, YOLO architectures have become popular because of the balance between speed and accuracy.

3. METHODS AND MATERIALS

In my project, we designed two dataset using roboflow. These two dataset are main dataset has 700 ship images 5 classes. Also, the other dataset is augmentations are implemented main dataset 1260 total images achieved. For model YOLOV8n Nano was chosen as architecture. This model important for small dataset create less memorization and low parameters. Furthermore, allows high FPS values in cameras.

Tablo 1 Dataset Classes

Carrier	0	Ship carrying huge boxes
Container Ship	1	Carries Bulk Cargo
RORO	2	Carries Vehicles
Tanker	3	Carries oil,gas,chemical
Tugboat	4	Vessels pull other ships

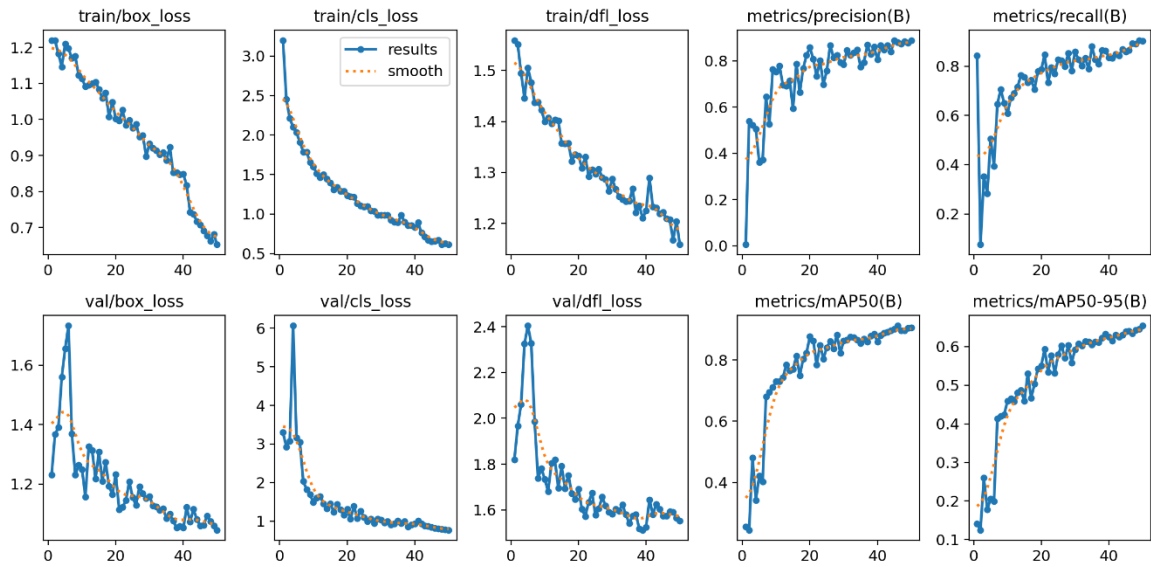
This classes are our dataset in small version we have 700 ship images to enable dataset to train model labeling process bounding box structures were done. After that data split to 70% train, validated (20%), and tested (10%) by taking the dataset using API Key in the Google Colab environment. In training process, as a model YOLOv8n, 50 epoch, image size 640x640, batch size 16 were used and model trained. After that performace metrics give us model success rate and best.pt file results.png, data.yaml file was exported and uploaded to github platform. To test our model a video upload Google colab environment 26 second long and 30 FPS and the model prediction confidence score saved. To sum up, videos give us augmentation step must be done.

Data augmentation operations follows Blur 1.5px, Horizontal Flip, Brightness 18%, Noise 1.2% reached 1260 images. These techniques were applied because of to prevent model memorize and force the model see different variations that is why model will succesfully recognize ships in different angles record. Blur applied because of camera moving wind field or weather conditions can make cameras blur so we prepare our model for this conditions. Dataset spit 840 images train set 67%, valid set 280 images 22% Test set 11% epoch 50.[8][9]

4. RESULTS AND DISCUSSION

Results are created by using yolov8 models and performance metrics used for give us evaluation. Also our model trained for 50 epochs.

Figure 4. 1. Results Output Image



In this result output image give us a chart showing training, validation, performance metric, mAP scores over 50 epochs.

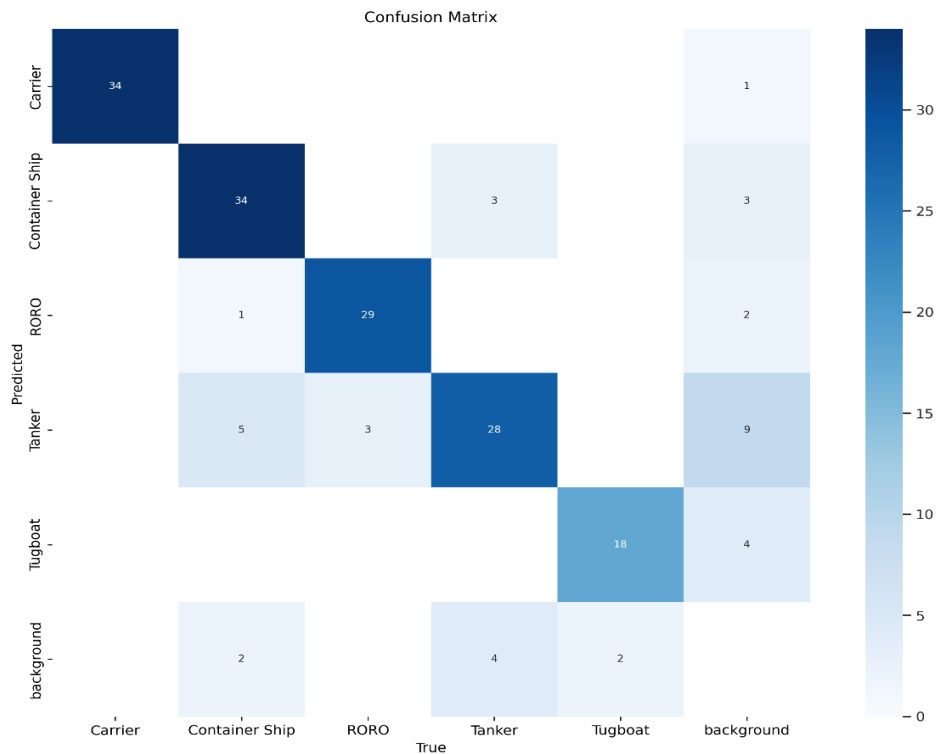
Metrics/Map50(B), The result we achieved in this graph was a success rate above 80% probably reach 90% so our model for ship object detection demonstrate high classification.

Metrics/precision(B), model scores increased in each epoch turn. Furthermore, when our model come the end of training precision score rate above %80 and with that we can measures how many ships model successfully recognized. Lastly, high precision performance metric give us less false positives can be happen but mostly accurately identified objects.

Val is used for validation in each epoch model loss values keep getting low that show us model does not have overfitting problem.

To summarize, our model correctly classifies ships and identify ship classes so we it seems Yolov8 model shows high success. So, model has been successfully trained.

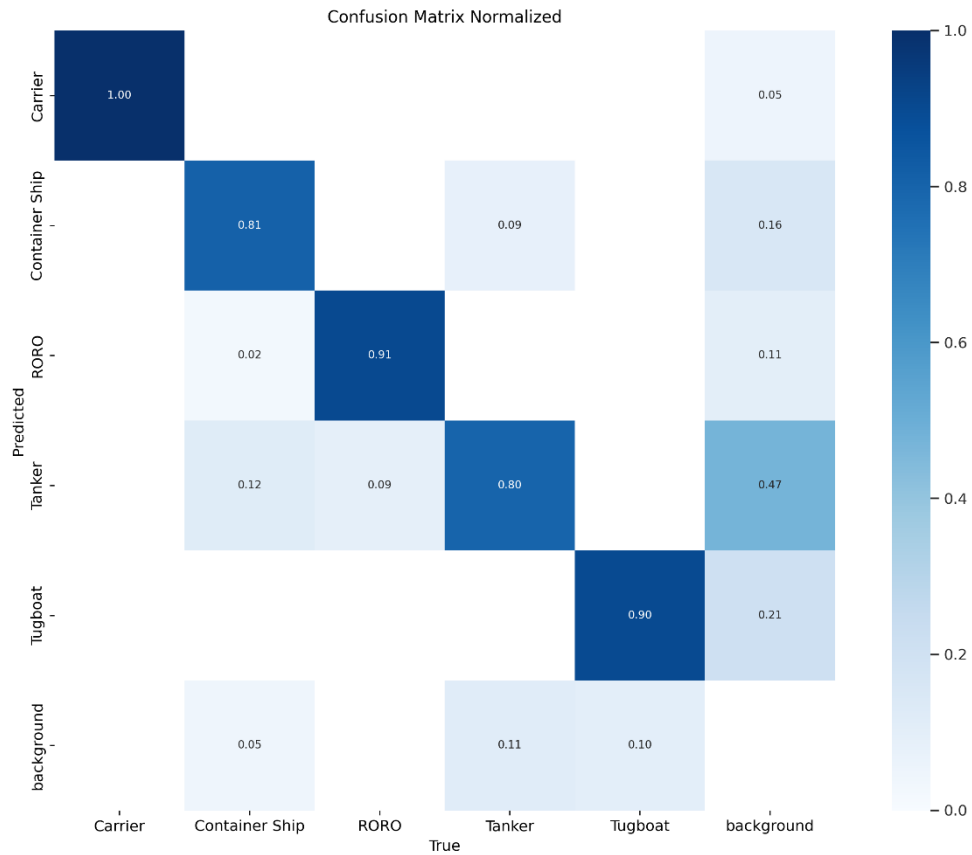
Figure 4. 2 Confusion Matrix



In this image show us Confusion Matrix meaning our model's classification performance, class confusion and vulnerabilities against background noise. Numerical values are validation that model predict accurately objects. So, we see 6 class most accuracy results are Carrier and Container Ship data. This means that our model predict 34 Carrier value true. But 1 values mistake Carrier as a background. Container Ship, total validation value is 40 and 34 is predicted correct classification was achieved by our model. But made inaccuracy classification 3 container ship as Tanker and 3 more as background. RORO, total validation is 32 value 29 value predicted true. Tanker, total validation is 45 value and 28 value predicted true. Tugboat, model classification achieved 18 true predictions.

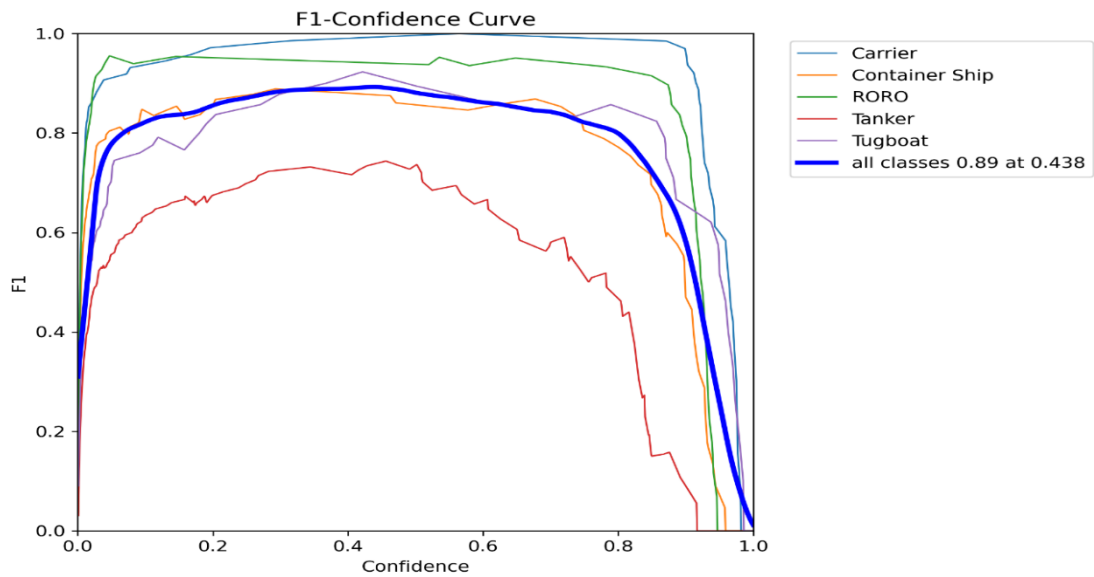
To summarize, Carrier and Container Ship classification achieved high prediction true results. Tanger class give us most class overlaps and FPA results that model made when doing prediction. Lastly, our dataset 700 value small size this create the lack of data resulting background noise classification. When agumentation done result will be change.

Figure 4. 3 Confusion Matrix Normalized



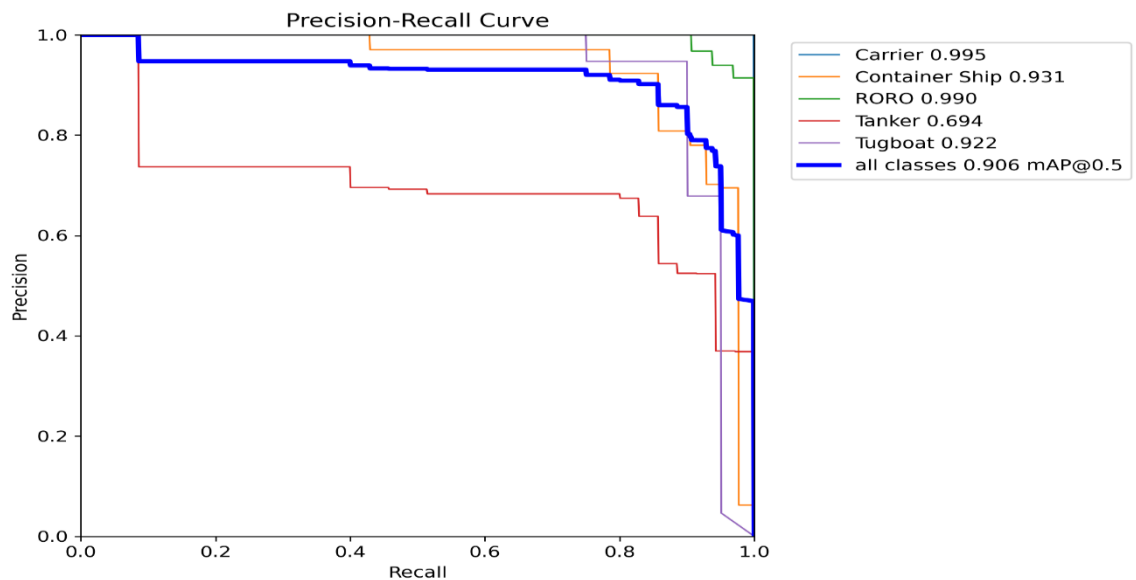
This graph is actually confusion matrix percentages version that called confusion matrix normalized it show us class ratio between 0 and 1. Predicted True class close to 1 values and color would be more darker tone. Overall, detecting the ship types is giving good performance. Carrier class 1.00 means 100% recall and RORO 91% recall. Additionally, 47% of the background noise was identified as being a Tanker and 21% as Tugboats, which indicates a weakness structurally in the background separation capability of the limited training dataset from which the model was created and that there was no additional data used to augment the dataset. So, tanker class must be focused on to lower background value.

Figure 4. 4 F1-Confidence Curve



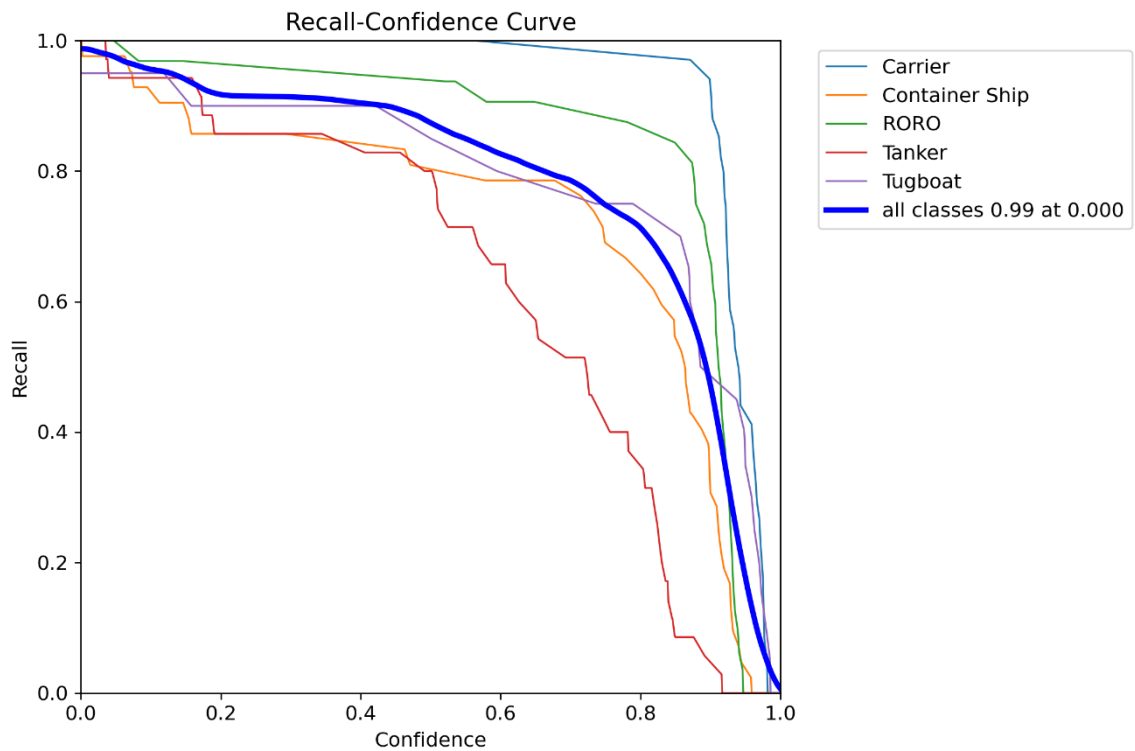
This F1-Confidence Curve Graph give us mean of Precision and Sensitivity that balances performance between false positivies and negative errors information[9].So, our model gained best confidence score inside all classes 0.89 at 0.483.This is important result for evaluating the false positivie and negative results of a system inn the operational field.

Figure 4. 5 Precision-Recall Curve



This Precision-Recall Curve show us the model achieved a 0.906 mAP50 means 90.6% precision at IoU of 0.5 all classes was successful caught object boundaries.From AUC side Carrier class achieved 99.5% result highest performance.After that RORO follows 99% in precision-recall curve.Tanker AUC falls this means false positives within model.

Figure 4. 6 Recall-Confidence Curve



The Recall-Confidence Curve is measure of model fails locate area objects False Negative percentage rate when confidence threshold is increased.

The graph give us general sensitivity with blue line include all classes reach maximum. But detection rate decreased the detection confidence threshold continued to increase past 0.50. It was also apparent during the class-based analysis that both the Carrier class open blue, and the RORO class show green line , maintained a flat curve through the 0.80 confidence threshold, indicating that these ship types have very specific and strong features that participate to the model.

However, decline of the curve for the Tanker class red line beginning around the 0.40 confidence threshold, shows that there are significant uncertainties associated with identifying this class. As a result because of tanker class we can say agumentation needed.

5. CONCLUSION

In this thesis, ship detection with yolov8 project explained. Ship data were obtained from roboflow and with creating API to call in Google Colab environment with train 50 epochs to detect 5 ship classes. This project aim to help marine vessels detection and classification to apply smart port systems. Furthermore, result are show us YOLOV8 main 700 dataset performs successfully in ship detection and classification. Achieved in this project was a success rate 90.6% of the mAP50(B) score.

However, augmentationed data that 1260 images have achieved lower results than main data. The metrics were like mAP50(B) score 86.6%. This means overfitting happened. Validations losses steady and this create bottleneck to happen. When we look other classes RORO and Carrier give us best result. But when we come to tanker and tugboat classes model gives structural collapse. As a result 700 data analysis give us best performance metrics.

The output:

Tablo 2 Augmentation Version

Class	mAP50	mAP50-95
Carrier	0.993	0.849
Container Ship	0.862	0.546
RORO	0.971	0.694
Tanker	0.658	0.328
Tugboat	0.847	0.572

Tablo 3 Main Model Version Output

Class	mAP50	mAP50-95
Carrier	0.995	0.898
Container Ship	0.931	0.657
RORO	0.99	0.737
Tanker	0.694	0.353
Tugboat	0.922	0.631

To sum up, this type project can be improve if the dataset has more permission to use real time camera data and diversified images. Furthermore maritime port integrated smart port system monitoring results and can get higher accuracy rates and improve this era.

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ÖZGEÇMİŞ



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İnsan Kaynakları Stajyeri- Zorunlu Staj

İşe alım süreçlerinde yer aldım ve adayların bilgilerini MS Excel tablolarına işledim aynı zamanda mülakat durumları gibi bilgileri güncelledim. Özlük dosyalarının güncel ve tam olması ile ilgilendim. SAP nedir ve bu sistem üzerinden süreç yönetimini öğrendim. Staj süresi boyunca toplam iki sunum yaptım ve son sunum staj süresince elde ettiğim kazanımlar ile ilgiliydi.

SCA SOCIAL

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Staj süresince gerekli proje yönetimi eğitimlerimi tamamladım ve katılım gösterdim. Staj sonu istenilen proje kapsamı içerisinde bu staj boyunca elde ettiğim teorik bilgilerimi pratiğe döktüm ve detaylı gantt şeması, organizasyon dosyası, proje başlatma belgesi, yapay zeka ve veri bilimi projelerini yapıp stajı başarılı bir şekilde bitirmiş bulunmaktayım.